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NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION

Question paper

Month and Year OF Exam: May, 2025
Duration: 3 Hrs.
Subject: Object Oriented Programming with JAVA

Programme: B.Tech 4th Sem
Max. Marks: 50
Code: IIPC 213

Note: Attempt any five questions.

Q1. An airline company maintains a unified reservation system. There are three types of passengers: Economy, Business, and First Class. All inherit from the base class Passenger. Fare calculation varies based on type.

- a. Explain how inheritance and polymorphism are applied to handle different passenger types using the same `calculateFare()` method. [5 Marks]
- b. Create an abstract class Passenger with abstract method `calculateFare()`. Implement this method in three subclasses. Show polymorphic behavior using an array of Passenger references. [5 Marks]

Q2. A logistics system handles both land and air deliveries. Each must implement a unified `DeliveryService` interface while hiding internal cost calculation logic.

- a. Explain how abstraction and interfaces together promote loose coupling and design flexibility. Give real-world analogy. [5 Marks]
- b. Create interface `DeliveryService` with method `calculateDeliveryCost()`. Implement it in `LandDelivery` and `AirDelivery`. Use an abstract class `BaseDelivery` to hold shared data. Demonstrate interface-based polymorphism. [5 Marks]

Q3. A smart home automation app controls devices like lights and ACs. Commands like `turnOn()` may differ per device and accept varied inputs (e.g., with/without time or mode).

- a. Discuss the differences between method overloading and overriding in the context of command execution. Why is each required here? [5 Marks]
- b. Create a base class `Device` with overloaded versions of `turnOn()` and override it in `Light` and `AC`. Show how both overloading and overriding coexist in a class hierarchy. [5 Marks]

Q4. A payment gateway validates credit card details before processing payments. Input like card number and CVV must be encapsulated and verified, with proper error handling.

- a. Explain how encapsulation secures sensitive data, and how exception handling ensures robust user interaction. [5 Marks]
- b. Create a class `Payment` with private fields for card number and CVV. Provide setters with validation logic, throwing a custom `InvalidCardException` on failure. Handle exception using try-catch-finally. [5 Marks]

Q5. A company wants to organize a document reader tool using proper Java packages and file input-output operations.

a. Describe how packages improve modularity and how Java I/O classes interact with the file system efficiently. [5 Marks]

b. Create a package docreader with class ReaderTool that reads a file and prints number of characters and lines. Use BufferedReader. Demonstrate use of package and import syntax clearly. [5 Marks]

Q6. An e-commerce system updates product inventory and processes orders concurrently. To avoid data corruption, proper synchronization is required.

a. Explain the importance of multithreading and synchronization in concurrent systems. What issues arise without it? [5 Marks]

b. Create a class Inventory with synchronized method updateStock(). Launch two threads: one reduces stock (purchase), one adds stock (restock). Use Runnable. Show synchronized output. [5 Marks]

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